

## **Water, Agriculture, and Decolonized Food Security in Ontario, Canada**

Kumary Chiquinquirá Ponnambalam, Thangaraj Draviam, Kumaraswamy Ponnambalam\*

\*Corresponding author at Department of Systems Design Engineering, University of Waterloo, Waterloo, Ontario, Canada, N2L 3G1. [Ponnu@uwaterloo.ca](mailto:Ponnu@uwaterloo.ca)

(Work presented as an invited talk at the International Conference on Water and Environmental Research, 10<sup>th</sup> ICWRER conference, December 9-11, 2025, Karunya University, Coimbatore, India)

### **Abstract**

The reality of changing systems—shaped by rapidly shifting climate patterns, weather extremes, and migration flows—is difficult to calculate, particularly as large environmental NGOs and western philanthropy, even when operating with good intentions, continue to displace the true keepers and restorers of ecological systems. This paper rethinks food sustainability with water and community as the foundation of food security and by examining how grassroots, bioregional, and anarchist (opposing all forms of hierarchical authority) informed movements can support more resilient and just approaches to environmental governance. In Ontario's agricultural landscapes—historically organized for market efficiency rather than ecological resilience—industrial water management practices and corporate-controlled greenhouse expansion have intensified food injustice, further alienating family farmers from traditional knowledge and water-efficient systems. We propose a decolonized (one group controlling most essential resources is disallowed) framework rooted in sustainable water governance, Indigenous and migrant agricultural knowledge, multicultural neighbourhood ecologies, and community-led design. Through analysis of rainwater harvesting, gravity-fed irrigation, agroecology, and local self-governance—combined with system dynamics and agent-based modeling—we show how decentralized water–food systems can stabilize regional economies while mitigating climate and economic risks. Case studies of now mostly destroyed Indigenous land trusts and current migrant-led farms demonstrate practical pathways for Ontario to anchor food sustainability in justice, bioregional stewardship, and community-driven innovation, including the potential role of technologically advanced neighbourhoods. We outline system dynamic and agent-based modelling to study current status of the neighbourhood and to study impacts of proposed changes towards a decentralized production of food in local communities with water (a major input affected by climate change) available in a sustainable manner.

### **1. Introduction: Systems of Change in a Time of Climatic Turbulence**

Climate change, extreme weather variability, and global migration present complex systemic challenges whose outcomes are deeply uncertain. These forces reshape land, identity, economies, and ecosystems simultaneously. While large environmental NGOs and philanthropy often dominate global climate discourse, they can unintentionally marginalize grassroots knowledge systems—particularly those of Indigenous communities, rural stewards, and culturally embedded

neighbourhoods. We note that unlike in the days of dominance before colonial days, colonial days brought the first large scale transfer of resources (both natural and man-made) which continues today. This large scale transfer of resources is now dominated by a few countries and corporations. The displaced local systems often hold the most profound understanding of ecological rhythms, seasonal transitions, and the cultural memory essential for long-term resilience. This paper examines how neighbourhood-level ecologies, Indigenous agricultural systems, and maple sugar bush forests reveal alternative paradigms of environmental stewardship in our community in Waterloo, Ontario, which is also not falling behind in technology and AI-development and use. The neighbourhood is also aligned with grassroots anarchist and bioregional traditions attempting to strengthen local forms of self-organization. Together, these perspectives offer insight into how communities can re-anchor environmental governance in local context, lived experience, and collective agency.

## **2. Neighbourhoods as Cultural-Ecological Knowledge Systems**

Neighbourhoods are not merely clusters of buildings; they are living cultural ecologies. Built environments—whether high-rise low-income housing or low-density subdivisions—generate distinct subcultures shaped by social, economic, and ecological conditions although high-rise communities need greater effort (the new developments along railway lines in Toronto waterfront is a good example) for community organization unlike others that have more natural tendencies to form communities. These subcultures are often subversive, healing, and adaptive. They produce local distinctiveness through dance, dialect, dress, movement, and ritual, helping communities respond to unique environmental pressures. Neighbourhoods also serve as informal mental health infrastructures which is a major cobenefit. Social connection, shared identity, and collective creativity help reduce anxiety, loneliness, and stress. Evidence shows that residents of supportive neighbourhood experience [ 1-2]:

- lower rates of depression and cardiovascular disease,
- higher levels of physical activity,
- increased resilience and coping capacity,
- stronger sense of belonging and purpose.

Neighbourhoods function as proto-commons—spaces where people negotiate shared resources, reinforce cultural memory, and build trust. They are vital governance units for environmental adaptation, offering a scale at which collective action can flourish. We use our neighbourhood as a case study to explain further.

## **3. Case Study: The Lakeshore Neighbourhood in Waterloo**

The Lakeshore neighbourhood in Waterloo, Ontario, Canada offers a microcosm of ecological and cultural integration. Figure 1 presents the actual map of the small central part of the neighbourhood and, in order to make modeling easier and understandable, a schematic automatically derived is presented in Figure 2. Later this schematic is used to develop the mathematical and computer models. It hosts a diverse multicultural population living across mixed-income housing. Its social infrastructure includes:

- Farmland
- Housing (individual, townhouse, low rise and high-rise buildings)
- Universities, college, and tech parks

- Lakes and wetlands
- Parks and trails
- Children's playgrounds
- Sports stadiums and playgrounds
- Schools, churches
- Libraries
- Walk-in clinics and pharmacies
- Gyms
- Meeting places for cultural events
- Proposed HUGE hospital (to be the second largest in Canada with multispecialities)
- Proposed Hotel, a large high-rise complex in addition to the many already existing, where students live close to the two major universities
- Farmer's market
- Two plazas filled with independent businesses including an Indian grocery store, German bakery, hair cutting salons of both western and eastern cultures, western music school among others.

These businesses reflect the cultural fusion of their owners and customers, forming a vibrant ecosystem of ideas and practices. The neighbourhood itself is embedded in a mixed-age deciduous sugar maple stand, inhabited by wildlife such as foxes, coyotes, small mammals, and numerous bird species. These creatures shape neighbourhood narratives, seasonal cycles, and children's learning landscapes. The presence of high-tech industries and recently AI developing companies in the neighbourhood originates from two universities nearby with one well known university founded on technical studies, and families whose livelihoods are connected to this sector. This includes a strong, supportive Indian community in the last five decades that enriches cultural and social exchanges which include many Indian origin faculty members and teachers in schools. Thus, Lakeshore illustrates how technological expertise, cultural pluralism, and ecological richness can co-evolve within a single local environment.

#### **4. Sugar Maple Bush Forests: Ecological, Cultural, and Economic Cobenefits**

Sugar maple forests provide remarkable ecosystem services that extend far beyond syrup production. They form a powerful example of bioregional infrastructure where culture, ecology, and economy intersect with water resources development and management. These forests form a large body of land for both aquifer replenishment and flood control.

##### **Environmental Cobenefits**

- **Carbon sequestration:** Maple forests store large amounts of carbon in biomass and soils. Our sugar bushes capture and store roughly eight times the carbon they emit during production. Natural capital valuations estimate these forests contribute approximately **\$20,000 per hectare per year**.
- **Biodiversity:** Sugar bushes support diverse flora and fauna, from pollinators to large mammals. Sustained forest management enhances resilience to pests and disease.
- **Air and water filtration:** Forests purify air and ensure steady supplies of clean drinking water.
- **Soil preservation:** Healthy forest cover prevents erosion and maintains moisture balance.

- **Climate resilience:** Mixed-age maple stands are more adaptable to environmental change than monoculture plantations.

#### **Economic and Cultural Cobenefits**

- **Rural livelihoods:** Many sugar bushes are family-run operations, central to rural economies.
- **Additional forest products:** Timber, berries, mushrooms, and medicinal plants provide diverse income streams.
- **Cultural and recreational value:** Maple forests support tourism, education, rituals, and seasonal celebrations.
- **Land conservation:** Economic viability from syrup production encourages forest preservation rather than conversion to development.

Maple forests thus represent multi-layered commons—ecologically productive, culturally meaningful, and economically vital.

### **5. Indigenous Agriculture Along the Grand River: Food Sovereignty and Cultural Continuity**

Indigenous food systems in the Grand River territory—particularly those of the Haudenosaunee—exemplify sustainable, culturally grounded ecological governance. The *Three Sisters system* (corn, beans, squash), known as *Tey’o’nhekwén* (“Our Sustenance”), embodies ecological harmony and community interdependence:

- Corn provides the vertical structure.
- Beans fix nitrogen in the soil.
- Squash shades and protects the ground.

These practices are embedded within *Traditional Ecological Knowledge (TEK)*, which includes water protection, wildlife respect, rotational harvesting, and ceremonial gratitude.

#### **Cultural and Spiritual Dimensions**

- Food is tied to ceremony, storytelling, and ancestral memory.
- Agriculture supports mental health, community cohesion, and spiritual well-being.
- TEK transmits moral principles of reciprocity, stewardship, and relationality.

#### **Economic Dimensions**

- Revitalized Indigenous agriculture supports food sovereignty and reduces dependency on industrial food systems.
- Community initiatives such as the Mohawk Seed keeper Gardens preserve heritage seeds and re-establish traditional food sources.
- The Six Nations Development Corporation is building agricultural ventures to create a sustainable “food basket” economy.

Indigenous agriculture offers a model of multi-generational resilience grounded in land-based ethics. It is to be noted that, unfortunately, during the invasion by Europeans into this North-American land the Indigenous agriculture was not widely understood as superior in sustainability to the large scale mono-culture followed by others such as in Europe and in Asia.

### **6. Neighbourhoods as Commons: Applying Elinor Ostrom’s Framework**

Elinor Ostrom's [3] research on the governance of the commons provides an essential lens for understanding neighbourhood-level resilience. Ostrom demonstrated that communities can self-organize to manage shared resources effectively without top-down control.

Key principles relevant to neighbourhood ecologies include:

- **Polycentric governance:** Multiple, overlapping centres of decision-making allow for flexible, context-appropriate solutions.
- **Self-organization:** Communities develop their own institutions—urban gardens, tool libraries, neighbourhood watches.
- **Monitoring and enforcement:** Local norms, rather than external policing, help sustain shared resources.
- **Graduated sanctions:** Gentle enforcement builds trust without undermining relationships.
- **Locally tailored rules:** One-size-fits-all environmental policies rarely match local ecological specificity or cultural values.
- **Building social trust:** Strong social ties produce higher collective action capacity.

Neighbourhoods, maple forests, Indigenous farms, and grassroots environmental networks all function as types of commons—complex socioecological systems requiring continuous negotiation and cooperation.

## 7. AI, Anarchism, and Bioregionalism: Toward a New Model of Community Governance

This project explored how AI communities, grounded in grassroots anarchist principles, might support bioregional systems of care. Anarchism emphasizes decentralized governance, mutual aid, and community self-determination—all principles aligned with Ostrom's vision of commons governance. Internet based social networks already have huge influence in changing opinions, electoral prospects, commercial successes, social practices, among others and are modern examples of anarchism.

AI tools, if developed in community-oriented ways, could:

- amplify local decision-making rather than replace it,
- map neighbourhood resources, wildlife flows, and seasonal changes,
- support Indigenous and local knowledge systems,
- enhance communication across diverse cultural groups,
- help residents monitor ecological indicators collaboratively,
- strengthen community associations in responding to climate impacts.

Instead of reinforcing top-down NGO structures, AI could help communities regain control over their ecological futures.

## 8. Systems Dynamics and Agent-Based modelling

To understand how neighbourhood-level governance, cultural activities, water management, and food-sharing systems interact over time, we employ a complementary modeling framework that integrates Agent-Based Modeling (ABM) and System Dynamics (SD). These models [4] operationalize the socio-ecological concepts discussed throughout the paper—commons governance, bioregional stewardship, anarchist-informed decentralization, and decolonized water–food sovereignty—within the spatial and cultural context of the Waterloo Lakeshore neighbourhood. See appendices for the details.

By integrating both modeling approaches, the neighbourhood gains a powerful decision-support tool aligned with decolonization, commons governance, and bioregionalism (Glossary):

- **Decolonization:**  
Testing policies that shift power from external corporations and NGOs to local residents, Indigenous land stewards, and migrant farmers.
- **Commons governance:**  
Understanding how community-managed projects (gardens, lakes, forests, food hubs) evolve and stabilize.
- **Bioregionalism:**  
Evaluating how Lakeshore's ecology—maple forests, wetlands, wildlife, microclimates—interacts with water and food systems over time.
- **Anarchist-informed decentralization:**  
Exploring how reducing hierarchical control enhances mutual aid, direct democracy, and adaptive self-organization.

Together, the ABM and SD frameworks allow Lakeshore residents, institutions, planners, and researchers to test:

- water conservation strategies,
- Indigenous and agroecological farming systems,
- community food hubs and virtual markets,
- cultural and health events,
- governance reforms,
- urban design choices (parks, trails, plazas),
- AI-enabled monitoring tools,
- ecological restoration of maple forests and wetlands.

This modeling integration ensures that policy interventions strengthen local autonomy, cultural vitality, ecological resilience, and food–water sovereignty—all essential elements of a decolonized neighbourhood future.

## 9. Conclusion: Reclaiming Environmental Futures Through Local Ecologies

Systems of change—climate disruption, migration, biodiversity loss—cannot be understood or governed solely through centralized institutions. Real resilience emerges from the ground up: through neighbourhood cultures, Indigenous agricultural traditions, maple forest ecologies, and community-led governance systems. The Lakeshore neighbourhood in Waterloo illustrates how cultural diversity, ecological richness, and technological capacity can coexist within a bioregional setting. Maple sugar bushes demonstrate how rural ecological communities sustain both biodiversity and livelihoods. Indigenous agricultural systems along the Grand River reveal a timeless model of reciprocity and sustainable stewardship. Ostrom's work offers a blueprint for neighbourhood-scale governance of shared resources. Together, these systems show that the future of climate adaptation lies not in displacing local stewards, but in empowering them—through community-based governance, ecological literacy, cultural revitalization, and the ethical use of AI building both social and cultural capital to compensate for the over emphasis on economic capital.

## Glossary

- *Anarchist* - Anarchists propose social and economic systems based on principles like mutual aid, solidarity, and direct democracy, often described as a blend of individual liberty and communal equality.
- *Decolonization* - Decolonization could be understood as “taking away the colonial” but this raises the question of what colonial means. Colonization involves one group taking control of the lands, resources, languages, cultures, and relationships of another group. Note that in these days the group controlling can also be a large corporation or other non-governmental entity.
- *Deciduous* - Most trees and shrubs fall into one of two categories: deciduous or coniferous. The deciduous trees have leaves that fall off yearly, tropical ones shed them throughout the year unlike ones in Ontario which shed in the Fall predominantly. Coniferous trees bear cones and have needles or scales that do not fall off.
- *Commons governance* - is the collective management and regulation of shared resources, such as natural resources or digital commons, by the communities that use them.
- *Bioregionalism* - advocacy of the belief that human activity should be largely restricted to distinct ecological and geographical regions.
- *Social capital* - the networks of relationships among people who live and work in a particular society, enabling that society to function effectively.
- *Cultural capital* - the non-financial assets such as skills, education, knowledge, and tastes that can be exchanged for social or economic advantages and upward mobility

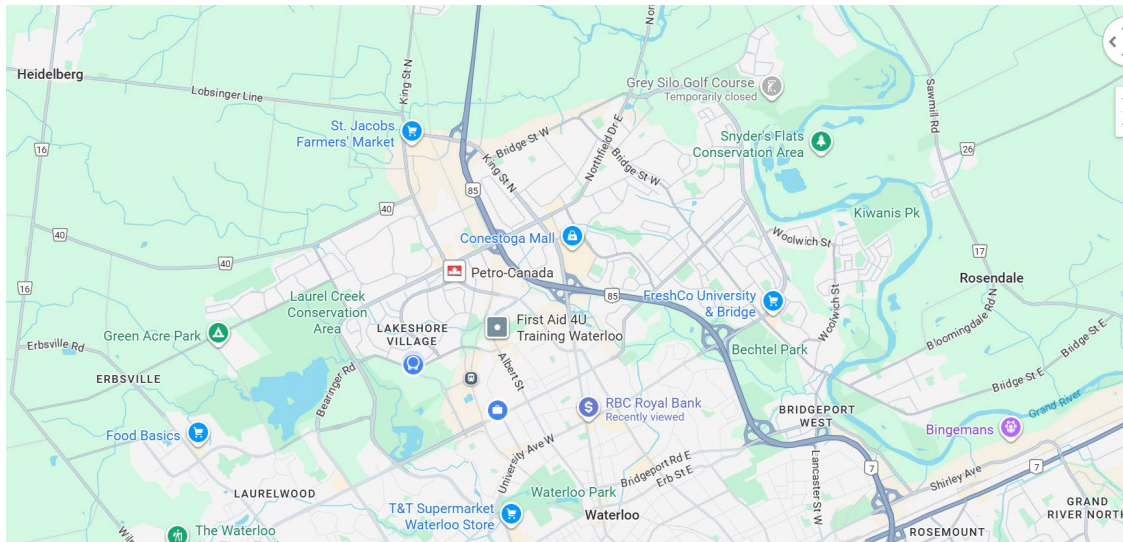


Figure 1. Google map of our Lakeshore neighbourhood, Waterloo, Ontario, Canada

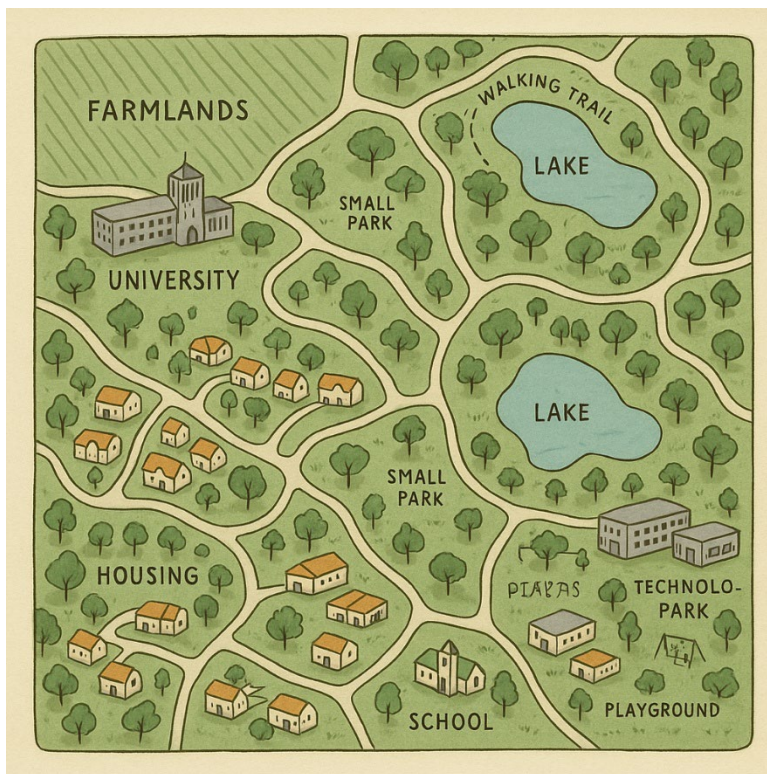


Figure 2. Schematic of our Lakeshore neighbourhood, Waterloo, Ontario, Canada



# Appendix A: Agent-Based Model Description for the Neighbourhood Water–Food–Commons System

## Purpose of the ABM

This model simulates how households, farmers, institutions, environmental features, and knowledge systems interact to produce neighbourhood-level outcomes in water storage, food production, green infrastructure, social and cultural capital, innovation, economic activity, and population.

## Environment (Spatial Landscape)

The model grid represents a stylized version of the neighbourhood map and includes:

See the list in section 3 of the article.

Each patch tracks land use, soil moisture, infiltration capacity, vegetation quality, and water runoff patterns.

## Agent Types

1. Household Agents (could have basic attributes of age, sex, education, ethnicity, income)
  - Attributes: water demand, food access, conservation behaviour, social propensity.
  - Behaviours: water use, consumption, participation in social interactions, migration decisions.
2. Farmer Agents
  - Attributes: farm size, irrigation type, agroecology adoption, labour.
  - Behaviours: irrigation, crop cultivation, soil stewardship, labour use, knowledge sharing.
3. Institution Agents
  - Types: schools, churches, plazas, libraries, medical clinics, shops, restaurants
  - Behaviours: host community events, increase social capital, spread conservation knowledge, food products availability, health care
4. Landscape / Environment Agents
  - Patches represent land uses, including parks, maple forests, lakes, tech parks.
  - Behaviours: infiltration, evaporation, water storage, biomass retention.
5. Animals
  - coyotes, deer, fox, rabbits, domestic cats and dogs
  - Behaviours: predator-prey interactions

## Agent Interactions

- Social interactions: households meet in schools, parks, plazas, increasing social capital.
- Economic interactions: households work in tech parks, farmers supply food to plazas.
- Ecological interactions: parks influence infiltration, farms affect water usage, lakes recharge soil moisture.
- Knowledge sharing: conservation knowledge spreads via institutions; agroecology spreads among farmers.

## Behavioural Rules Examples

### 1. Household Water Use:

$$\text{Water Use} = \text{base-use} * (1 - \text{conservation-knowledge})$$

### 2. Farmer Crop Yield:

$$\text{Yield} = f(\text{soil-water, green-infrastructure, social capital, agroecology-level})$$

### 3. Social Capital Growth:

$$C = C_0 + \alpha * \text{interactions} - \text{stress factors}$$

### 4. Migration:

Migration depends on neighbourhood attractiveness driven by food, green space, economy, and water stress.

### 5. Park and Forest Maintenance:

Green infrastructure improves with social capital and degrades with population pressure.

## System-Level Emergent Metrics

- Water storage: emerges from rainfall, infiltration, evaporation, withdrawals.
- Food production: sum of farmer yields.
- Population: number of households.
- Green infrastructure: average park/forest condition.
- Social capital: aggregate cohesion score.
- Knowledge capital: innovation from institutions and community.
- Economic capital: local income and job creation.

## ABM Implementation Summary

The ABM model includes:

- Patches representing spatial land-use cells.
- Households, farmers, and institutions as agents.
- Hydrology routines (rainfall, infiltration, evaporation).
- Behavioural decision-making (irrigation, water use, migration).
- Aggregate variables W, F, N, G, C, K, E aligned with the system dynamics model explained later.

## Role of Agent-Based Modeling (ABM)

The ABM represents the neighbourhood as a collection of interacting agents—households, farmers, institutions (schools, churches, plazas, libraries, medical clinics), environmental patches (farmland, parks, lakes, maple forests), and animals—each with distinct attributes and behavioural rules. This approach captures heterogeneity and local decision-making, making it particularly suitable for evaluating policies that depend on human behaviour, cultural norms, and ecological feedbacks. Because ABMs simulate actions and interactions at the micro scale, they allow the exploration of policies that emerge from lived experiences and neighbourhood-level cultural dynamics, such as:

## **1. Governance reforms and decentralized decision-making**

- Polycentric governance structures inspired by Ostrom's commons framework can be tested by modifying:
  - neighbourhood councils,
  - community-run water cooperatives,
  - participatory budgeting for green infrastructure, and
  - mutual aid networks (reflecting anarchist principles of non-hierarchical organization).
- The ABM shows how such governance shifts alter collective action, resource flows, and trust (social capital C).

## **2. Cultural and social events**

Since the neighbourhood includes libraries, community halls, churches, plazas, sports fields, and cultural meeting spaces (Section 3 and Appendix B), the ABM can simulate how:

- festivals,
  - communal meals,
  - arts gatherings,
  - cultural teaching circles, and
  - farmer's market events
- increase social cohesion (C), transmit knowledge (e.g., conservation, agroecology), reduce social stress, and strengthen neighbourhood proto-commons.

## **3. Health-related programs: walking and biking systems**

The Lakeshore neighbourhood's many **walking trails, parks, playgrounds, sports fields, gyms,** and its supportive terrain (Figures 1 and 2) make it ideal for studying health promotion.

The ABM can test the effects of:

- new bike lanes,
- enhanced trail connectivity,
- walk-to-school programs,
- cross-cultural walking groups,

by dynamically representing changes in:

- physical activity,
- social interaction,
- mental health (via reduced stress), and
- feedbacks to green infrastructure quality (G).

## **4. Local food production and sharing**

The model captures farmer agents, home gardens, food access, and institutional food availability.

Policy scenarios include:

- community-supported agriculture (CSA) networks,
- neighbourhood seed-sharing and agroecology training,
- decentralized food hubs and farmer's markets,
- coordinated food-sharing apps or virtual commons platforms,
- subsidies for water-efficient crops and Indigenous polycultures.

The ABM shows how these influence:

- food production (F),
- economic capital (E),
- household food access,
- migration and household stability (N), and

- the resilience of neighbourhood supply chains.

### Why ABM is crucial

ABM is uniquely capable of representing:

- agents are self-governing and adaptive bodies
- diverse cultural backgrounds,
- varying conservation behaviours,
- institution-specific effects,
- the spatial layout of the Lakeshore area,
- predator–prey interactions with local wildlife (foxes, coyotes, deer), and
- emergent patterns such as urban heat islands, food deserts, or social fragmentation.

These complex micro-to-macro relationships cannot be directly observed through aggregated SD models alone, explained next.

## Appendix B: Neighbourhood System Dynamics Model: Components, Causal Loops, and Differential Equations

### 1. Main System Components (Stocks)

W(t) – Water storage (lakes & aquifers)

F(t) – Local food & biomass production

N(t) – Population

G(t) – Green infrastructure quality

C(t) – Social capital / community cohesion

K(t) – Knowledge & innovation (university, tech parks)

E(t) – Local economic capital

U(t) – Knowledge specific to universities

T(t) – Practical knowledge from trade schools

### 2. Key Causal Loops

R1 – Regenerative water–food–economy loop

B1 – Water stress from population growth

R2 – Social capital & neighbourhood commons loop

R3 – Knowledge–economy–infrastructure loop

### 3. System Interactions

$W \rightarrow F \rightarrow E \rightarrow G \rightarrow W$

$G \rightarrow C \rightarrow G$

$C \rightarrow F$

$K \rightarrow E \rightarrow G \rightarrow W$

$N \rightarrow \text{Water Demand} \rightarrow W (-)$

$G \rightarrow \text{Amenity} \rightarrow \text{Attractiveness} \rightarrow N$

## 4. Differential Equations Modelling the Interactions

### 4.1 Water Storage $W(t)$

$$dW/dt = R_0 + \alpha G - \beta F \cdot F - \beta N - \beta E - \kappa W$$

### 4.2 Food Production $F(t)$

$$dF/dt = \eta W \cdot f(W) \cdot A F \cdot (1 + \eta G)(1 + \eta C) - \delta F$$

### 4.3 Population $N(t)$

$$dN/dt = \lambda A \cdot A_{attr} - \mu N \cdot N$$

$$A_{attr} = \omega_1 E + \omega_2 G + \omega_3 C + \omega_4 F$$

### 4.4 Green Infrastructure $G(t)$

$$dG/dt = \rho E + \rho C - \rho N - \kappa G$$

### 4.5 Social Capital $C(t)$

$$dC/dt = \sigma G + \sigma I \cdot I_{inst} - \sigma_{stress} \cdot (\theta_1 (N/(G+\epsilon)) - \theta_2 F) - \kappa C$$

### 4.6 Knowledge & Innovation $K(t)$

$$dK/dt = \phi U + \phi T + \phi C - \kappa K$$

### 4.7 Economic Capital $E(t)$

$$dE/dt = \chi F + \chi K + \chi N - \chi_{leak} \cdot E$$

While the ABM handles local heterogeneity and spatial interactions, the SD model provides a parallel macroscale understanding of the system. It tracks the co-evolution of seven key stocks. Using differential equations, SD exposes the dominant feedback loops—regenerative (R1, R2, R3) and balancing (B1)—driving long-term neighbourhood dynamics.

## Why SD is needed alongside ABM

- **Computational efficiency:**  
SD simulations run extremely fast, allowing rapid exploration of long-term policies such as water rights reforms or green infrastructure investment.
- **High-level policy analysis:**  
SD helps policymakers or community councils anticipate:
  - system-wide tipping points,

- unintended consequences (e.g., gentrification through economic expansion),
  - overall trajectories (sustainable, unstable, or inequitable growth).
- **Calibration and validation for the ABM:**  
ABM outputs (W, F, N, G, C, K, E) can be aggregated and compared with SD trajectories for robustness and policy convergence.
- **Scenario exploration:**  
SD can run hundreds of scenarios (e.g., drought cycles, demographic change) quickly before implementing detailed ABM simulations.

## Appendix C: Summary of Comparison of ABM and SD

| Feature                            | Agent-Based Modeling (ABM)                                                                                                                                                                                            | System Dynamics (SD)                                                                                                                                                                             |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Approach</b>                    | Bottom-up (starts from individuals)                                                                                                                                                                                   | Top-down (starts from the overall system)                                                                                                                                                        |
| <b>Focus</b>                       | Micro-level individual decisions, heterogeneity, and interactions                                                                                                                                                     | Macro-level aggregate flows, trends, and feedback loops (stocks and flows)                                                                                                                       |
| <b>Agents</b>                      | Agents are active, autonomous decision-makers with individual characteristics and rules                                                                                                                               | Components are passive homogeneous entities within stocks                                                                                                                                        |
| <b>Emergence</b>                   | Excels at capturing emergent phenomena resulting from local interactions                                                                                                                                              | Focuses on system structure and its effect on dynamics; system-level properties are more explicit                                                                                                |
| <b>Data Needs</b>                  | Requires detailed data at the individual level, which can be extensive                                                                                                                                                | Relies on aggregate data and quantified relationships between variables                                                                                                                          |
| <b>Computational Needs</b>         | Often demands greater computational resources, especially for large populations over long periods                                                                                                                     | Generally less computationally intensive for macro-level analysis                                                                                                                                |
| <b>Use Case for Neighbourhoods</b> | Ideal for simulating social interactions, pedestrian movement, individual energy use behavior, and how peer influence affects outcomes (e.g., adoption of sustainable practices or neighborhood segregation patterns) | Better suited for modeling resource management, overall population trends, and the impact of system-wide policies and incentives (e.g., housing supply-demand mismatch or waste flow management) |

## REFERENCES

- [1] Kepper, M.M., Myers, C.A., Denstel, K.D. *et al.* The neighborhood social environment and physical activity: a systematic scoping review. *Int J Behav Nutr Phys Act* **16**, 124 (2019). <https://doi.org/10.1186/s12966-019-0873-7>
- [2] [https://bchealthycommunities.ca/wp-content/uploads/2024/03/CYMW\\_Social\\_2.pdf?](https://bchealthycommunities.ca/wp-content/uploads/2024/03/CYMW_Social_2.pdf?)
- [3] Wilson, D.S., Ostrom, E., Cox, M.E. Generalizing the core design principles for the efficacy of groups, *Journal of Economic Behavior & Organization*, Volume 90, Supplement, 2013, Pages S21-S32, <https://doi.org/10.1016/j.jebo.2012.12.010>.
- [4]. Cassidy, R., Singh, N.S., Schiratti, PR. *et al.* Mathematical modelling for health systems research: a systematic review of system dynamics and agent-based models. *BMC Health Serv Res* **19**, 845 (2019). <https://doi.org/10.1186/s12913-019-4627-7>